SQL PORTFOLIO PROJECT | POSTGRESQL

Restaurant Business Analysis

Exploring restaurant sales, menu performance, customer behavior and outlet operations using SQL.



DATABASE
PostgreSQL

DATASET
6 relational tables

CORE SKILLS
JOINs, Aggregations, CTEs, Window Functions

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GitHub Repository:
Restaurant SQL Project

Project Overview

Analyzing restaurant data to uncover insights on sales, customer behavior, menu performance, outlet operations and reviews using SQL.

Objective

Analyze restaurant data to understand sales, orders, menu demand, customers, delivery performance and reviews.

Dataset

6 relational tables

customers | orders | order_items
outlets | menu_items | reviews

Tools

PostgreSQL | pgAdmin | GitHub

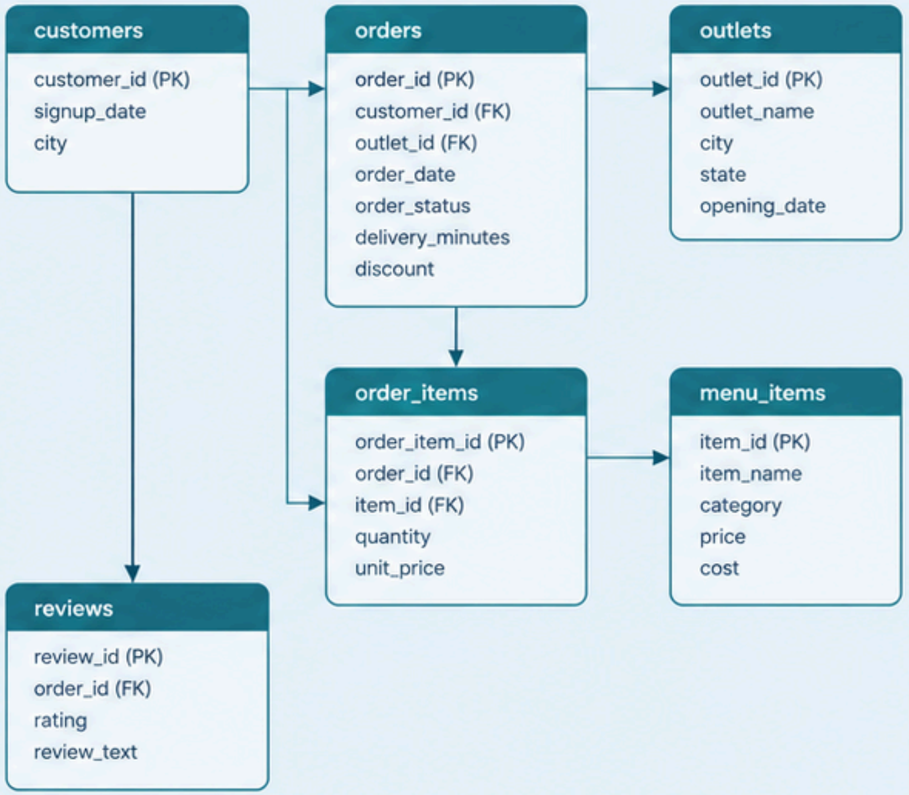
Dataset Tables

01	customers	Customer information
02	orders	Order details, dates, status, outlet and delivery information
03	order_items	Ordered items, quantities and unit prices
04	outlets	Outlet information and city
05	menu_items	Menu items and categories
06	reviews	Customer feedback and ratings

SQL Skills Demonstrated

SELECT | JOIN | GROUP BY | HAVING | AGGREGATE FUNCTIONS | CTEs | SUBQUERIES | WINDOW FUNCTIONS

Database Schema



01 | Find the total number of orders received by each outlet.

	outlet_name character varying (100)	count bigint
1	SpiceRoute Pune 1	519
2	SpiceRoute Delhi 1	516
3	SpiceRoute Bengaluru 2	507
4	SpiceRoute Mumbai 2	504
5	SpiceRoute Delhi 2	502
6	SpiceRoute Jaipur 2	502
7	SpiceRoute Jaipur 1	501
8	SpiceRoute Bengaluru 1	494
9	SpiceRoute Pune 2	481
10	SpiceRoute Mumbai 1	474

KEY OBSERVATION

- Highest Orders: SpiceRoute Pune 1 recorded the most orders.
- Lowest Orders: SpiceRoute Mumbai 1 recorded the fewest orders.
- Business Insight: Order volume varies across outlets, helping identify locations for improvement.

BUSINESS QUESTION

Display the outlet name and order count, sorted from highest to lowest.

ANALYSIS OBJECTIVE

To compare the total number of orders received by each outlet and identify the busiest outlets based on order volume.

SQL FILE

postgreSQL-01

02 | Best-selling menu items

Data Output Messages Notifications

	item_id integer	item_name character varying (100)	order_placed bigint
1	24	Masala Chai	441
2	30	Chilli Chicken	436
3	13	Dal Makhani Combo	433
4	21	Chocolate Shake	428
5	27	Chicken Wrap	428

KEY OBSERVATION

- Top-Selling Item: Masala Chai had the highest quantity sold.
- Sales Distribution: The top five items accounted for varying quantities of sales.
- Business Insight: Best-selling items can help optimize inventory planning and menu strategies.

BUSINESS QUESTION

Find the top 5 menu items by total quantity sold. Count only items belonging to delivered orders.

ANALYSIS OBJECTIVE

To identify the top five best-selling menu items by analyzing the total quantities sold through successfully delivered orders.

SQL FILE

postgreSQL-02

03 | Revenue by city

Data Output | Messages | Notifications

	city character varying (15)	revenue numeric
1	Bengaluru	424473.00
2	Pune	419039.00
3	Jaipur	412796.00
4	Delhi	405287.00
5	Mumbai	388468.00

KEY OBSERVATION

- Highest Revenue: Bengaluru generated the highest gross sales.
- Lowest Revenue: Mumbai recorded the lowest gross sales.
- Business Insight: Revenue differences help identify stronger markets and potential growth opportunities.

BUSINESS QUESTION

Calculate gross item sales for each city using delivered orders only. Display the city and total sales.

ANALYSIS OBJECTIVE

To calculate and compare gross item sales across cities using delivered orders only.

SQL FILE

postgreSQL-03

04 | Customer review analysis

	outlet_name character varying (100)	avg_rating numeric
1	SpiceRoute Pune 1	3.81
2	SpiceRoute Mumbai 2	3.75
3	SpiceRoute Jaipur 2	3.74
4	SpiceRoute Delhi 1	3.72
5	SpiceRoute Pune 2	3.72
6	SpiceRoute Bengaluru 2	3.72
7	SpiceRoute Delhi 2	3.71
8	SpiceRoute Bengaluru 1	3.69
9	SpiceRoute Mumbai 1	3.62
10	SpiceRoute Jaipur 1	3.60

KEY OBSERVATION

- Highest Rating: SpiceRoute Pune 1 received the highest average rating.
- Lowest Rating: SpiceRoute Jaipur 1 received the lowest average rating.
- Business Insight: Ratings help identify outlets that deliver better customer experiences and those needing improvement.

BUSINESS QUESTION

Calculate the average review rating for each outlet. Show the outlet name and average rating, highest rated first.

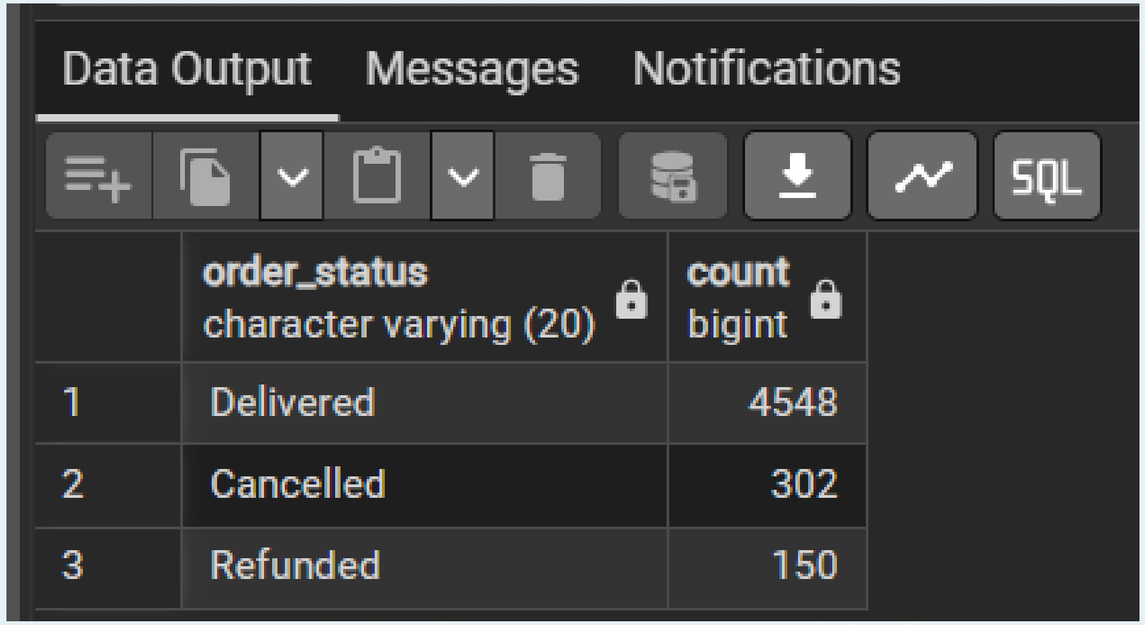
ANALYSIS OBJECTIVE

To calculate and compare the average customer review rating for each outlet.

SQL FILE

postgreSQL-04

05 | Orders by status



	order_status character varying (20)	count bigint
1	Delivered	4548
2	Cancelled	302
3	Refunded	150

KEY OBSERVATION

- Delivered Orders: 4548 orders were successfully delivered.
- Cancelled Orders: 302 orders were cancelled.
- Refunded Orders: 150 orders were refunded.

BUSINESS QUESTION

Count delivered, cancelled, and refunded orders separately. Display the status and count.

ANALYSIS OBJECTIVE

To count orders in each status and understand the overall order distribution.

SQL FILE

postgreSQL-05

06 | Average order value by city

Data Output | Messages | Notifications

	city character varying (15)	average_order_value numeric
1	Mumbai	456.32
2	Pune	474.03
3	Jaipur	467.28
4	Delhi	453.18
5	Bengaluru	482.79

KEY OBSERVATION

- Highest AOV: Bengaluru had the highest average order value.
- Lowest AOV: Mumbai recorded the lowest average order value.
- Business Insight: Differences in average order value reveal variations in customer spending across cities.

BUSINESS QUESTION

Calculate average order value for each city using delivered orders.

ANALYSIS OBJECTIVE

To calculate and compare the average value of delivered orders across cities.

SQL FILE

postgreSQL-06

07 | Outlet cancellation rate

	outlet_name character varying (100)	total_orders bigint	cancelled_orders bigint	cancelled_rate numeric
1	SpiceRoute Mumbai 2	504	37	7.34
2	SpiceRoute Delhi 1	516	37	7.17
3	SpiceRoute Mumbai 1	474	29	6.12
4	SpiceRoute Pune 2	481	29	6.03
5	SpiceRoute Jaipur 1	501	30	5.99
6	SpiceRoute Bengaluru 1	494	29	5.87
7	SpiceRoute Pune 1	519	30	5.78
8	SpiceRoute Bengaluru 2	507	29	5.72
9	SpiceRoute Delhi 2	502	26	5.18
10	SpiceRoute Jaipur 2	502	26	5.18

KEY OBSERVATION

- Highest Cancellation Rate: SpiceRoute Mumbai 2 and SpiceRoute Delhi 1 have the highest cancellation percentage.
- Lowest Cancellation Rate: SpiceRoute Jaipur 2 and SpiceRoute Delhi 2 have the lowest cancellation percentage.

BUSINESS QUESTION

For each outlet, calculate total orders, cancelled orders, and cancellation percentage. Sort outlets by cancellation rate.

ANALYSIS OBJECTIVE

To compare total orders, cancelled orders, and cancellation percentages across outlets.

SQL FILE

postgreSQL-07

08 | Repeat customers

	customer_id [PK] integer	city character varying (15)	delivered_orders bigint
1	413	Pune	18
2	478	Pune	17
3	112	Jaipur	17
4	24	Delhi	16
5	230	Mumbai	16
6	73	Bengaluru	16
7	353	Jaipur	16
8	497	Mumbai	16
9	349	Bengaluru	16
10	110	Pune	15
11	380	Mumbai	15

KEY OBSERVATION

- Repeat Customers: 478 customers placed at least five delivered orders.
- Most Frequent Customer: Customer ID 413 had the highest number of delivered orders.
- Business Insight: Repeat customers indicate customer loyalty and opportunities for retention strategies.

BUSINESS QUESTION

Find customers who have placed at least 5 delivered orders. Show customer ID, city, and number of delivered orders.

ANALYSIS OBJECTIVE

To identify repeat customers by counting their successfully delivered orders.

SQL FILE

postgreSQL-08

09 | Popular categories by city

Data Output | Messages | Notifications

	city character varying (15)	popular_category character varying (50)	item_delivered bigint
1	Bengaluru	Beverages	297
2	Delhi	Beverages	312
3	Jaipur	Beverages	346
4	Mumbai	Beverages	327
5	Pune	Beverages	369

KEY OBSERVATION

- Most Popular Category: Beverages ranked highest in all five cities.
- Highest Sales: Pune recorded the highest quantity sold, with 369 items.
- Lowest Sales: Bengaluru had the lowest quantity sold, with 297 items.

BUSINESS QUESTION

Find the most popular menu category in each city based on quantity sold in delivered orders.

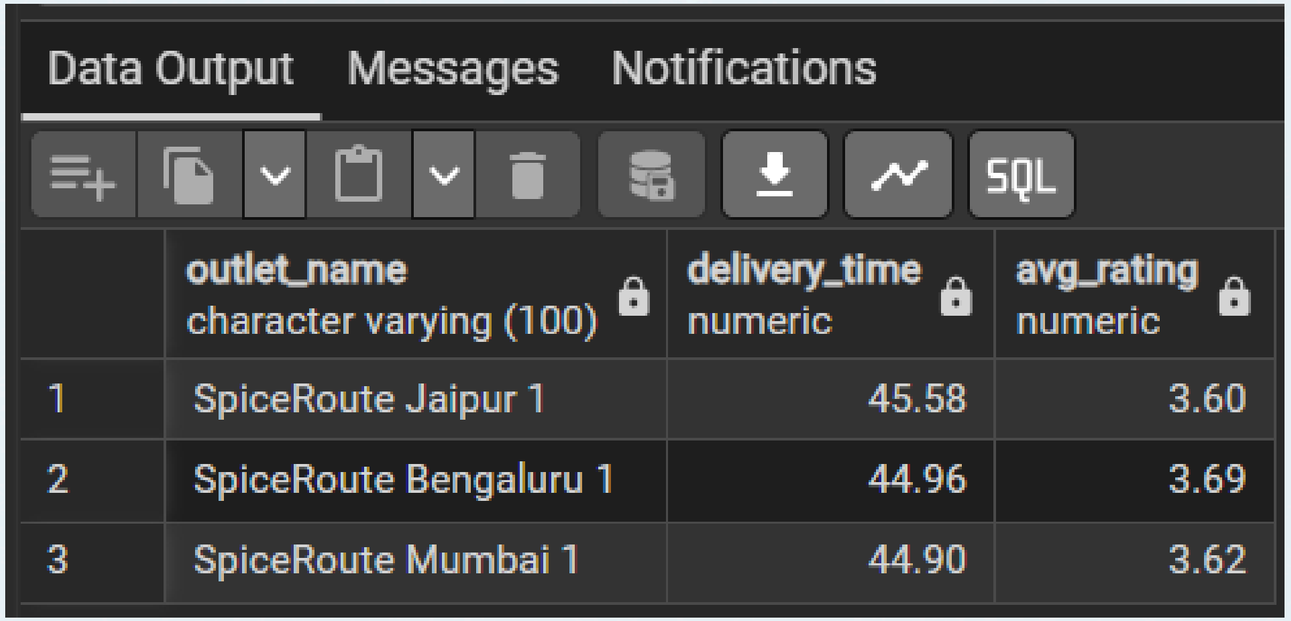
ANALYSIS OBJECTIVE

To identify the best-selling menu category in each city using delivered orders only.

SQL FILE

postgreSQL-09

10 | Delivery time and ratings



	outlet_name character varying (100)	delivery_time numeric	avg_rating numeric
1	SpiceRoute Jaipur 1	45.58	3.60
2	SpiceRoute Bengaluru 1	44.96	3.69
3	SpiceRoute Mumbai 1	44.90	3.62

KEY OBSERVATION

- Slowest Delivery: SpiceRoute Jaipur 1 had the highest average delivery time at 45.58 minutes.
- Lowest Rating: SpiceRoute Jaipur 1 received the lowest average rating of 3.60.
- Business Insight: SpiceRoute Jaipur 1 should prioritize improving delivery efficiency and customer satisfaction.

BUSINESS QUESTION

Compare average delivery time and average review rating for each outlet. Identify outlets that have slower delivery and lower ratings.

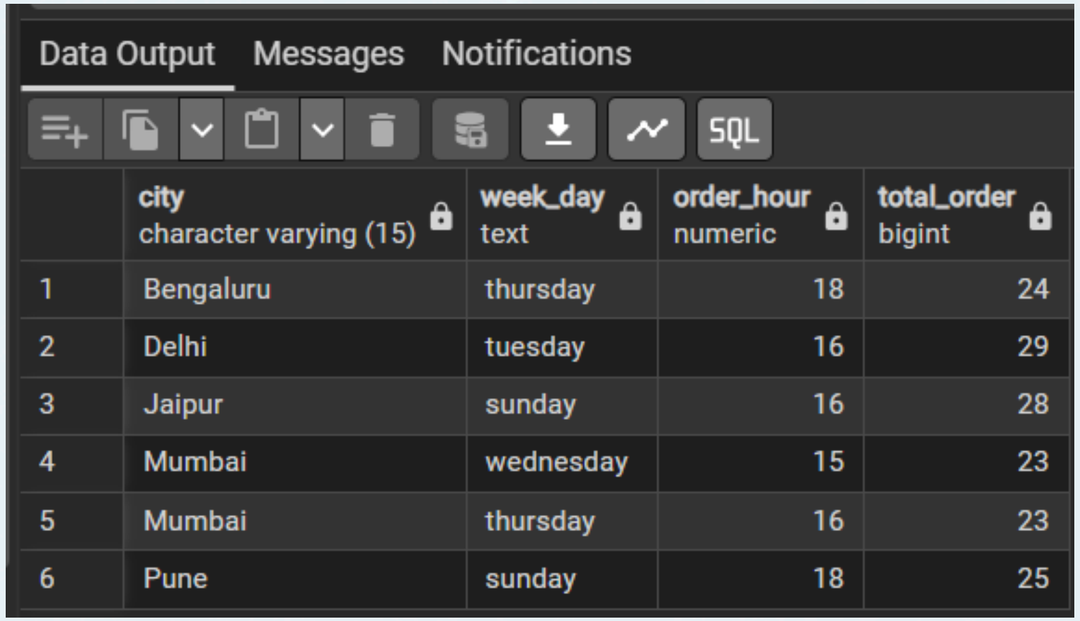
ANALYSIS OBJECTIVE

To compare average delivery time and average customer ratings across outlets.

SQL FILE

postgreSQL-10

11 | Peak ordering times



	city character varying (15)	week_day text	order_hour numeric	total_order bigint
1	Bengaluru	thursday	18	24
2	Delhi	tuesday	16	29
3	Jaipur	sunday	16	28
4	Mumbai	wednesday	15	23
5	Mumbai	thursday	16	23
6	Pune	sunday	18	25

KEY OBSERVATION

- Busiest Period: Sunday at 4 PM in Jaipur and Tuesday at 4 PM in Delhi recorded 28 and 29 orders, respectively.
- City Differences: Peak ordering times varied across cities, with Bengaluru peaking Thursday at 6 PM and Pune Sunday at 6 PM.

BUSINESS QUESTION

Find the busiest day of the week and hour of the day based on order count. Compare patterns across cities if possible.

ANALYSIS OBJECTIVE

To identify peak ordering periods and compare ordering patterns across cities.

SQL FILE

postgreSQL-11

12 | Month-over-month sales growth

	months text	gross_sales numeric	monthly_growth numeric	growth_percentage numeric
2	NOVEMBER 2025	171160.00	13114.00	8.30
3	DECEMBER 2025	155797.00	-15363.00	-8.98
4	JANUARY 2026	181156.00	25359.00	16.28
5	FEBRUARY 2026	160680.00	-20476.00	-11.30
6	MARCH 2026	183281.00	22601.00	14.07
7	APRIL 2026	157180.00	-26101.00	-14.24
8	MAY 2026	200950.00	43770.00	27.85
9	JUNE 2026	174526.00	-26424.00	-13.15
10	JULY 2026	181128.00	6602.00	3.78
11	AUGUST 2026	165212.00	-15916.00	-8.79
12	SEPTEMBER 20...	160947.00	-4265.00	-2.58

KEY OBSERVATION

- Highest Sales: May 2026 recorded the highest monthly gross sales.
- Sales Growth: May 2026 showed the highest month-over-month growth.
- Business Insight: Monthly sales trends help identify growth patterns and periods of declining sales.

BUSINESS QUESTION

Calculate monthly gross item sales and the percentage change compared with the previous month.

ANALYSIS OBJECTIVE

To calculate monthly gross item sales and evaluate month-over-month percentage changes to identify sales growth and decline trends.

SQL FILE

postgreSQL-12

13 | Top outlet in every city

Data Output | Messages | Notifications

	city character varying (15)	outlet character varying (100)	gross_sales numeric
1	Bengaluru	SpiceRoute Bengaluru 2	233400.00
2	Delhi	SpiceRoute Delhi 2	222663.00
3	Jaipur	SpiceRoute Jaipur 1	228303.00
4	Mumbai	SpiceRoute Mumbai 2	219683.00
5	Pune	SpiceRoute Pune 1	236747.00

KEY OBSERVATION

- Highest Gross Sales: SpiceRoute Pune 1 led its city with gross sales of ₹236,747.
- City Leaders: Each city had a top-performing outlet, with different gross sales across locations.
- Business Insight: Successful outlet strategies can be studied and applied to improve sales at other locations.

BUSINESS QUESTION

Rank outlets within each city by gross item sales and return the top-performing outlet in each city. Include ties if possible.

ANALYSIS OBJECTIVE

To rank outlets within each city by gross item sales and identify the top-performing outlet, including ties.

SQL FILE

postgreSQL-13

14 | Estimated menu-item margin

	item_name character varying (100)	quantity_sold bigint	sales numeric	estimated_cost numeric	estimated_item_margin numeric
1	Farmhouse Pizza	402	137908.00	61908.00	76000.00
2	Tandoori Paneer Pizza	379	137661.00	62535.00	75126.00
3	Paneer Butter Masala Com...	423	115287.00	51606.00	63681.00
4	Chicken Curry Combo	391	114409.00	54349.00	60060.00
5	Margherita Pizza	406	99074.00	42630.00	56444.00
6	Dal Makhani Combo	433	96267.00	42434.00	53833.00
7	Chilli Chicken	436	97364.00	44908.00	52456.00
8	Chicken Biryani	370	89880.00	41440.00	48440.00
9	Chicken Wrap	428	78592.00	32100.00	46492.00
10	Veg Biryani	426	78294.00	32802.00	45492.00

KEY OBSERVATION

- Highest Sales: Farmhouse Pizza generated the highest sales revenue.
- Highest Margin: Farmhouse Pizza achieved the highest estimated item margin.
- Business Insight: Comparing sales and margins helps identify profitable items and opportunities to optimize the menu.

BUSINESS QUESTION

For each menu item, calculate quantity sold, sales, estimated item cost, and estimated item margin. Only include delivered orders.

ANALYSIS OBJECTIVE

To evaluate each menu item's sales and estimated profitability using delivered orders only.

SQL FILE

postgreSQL-14

15 | Executive performance report

	outlet_name character varying (100)	delivered_orders bigint	gross_item_sales numeric	avg_delivery_time numeric	avg_rating numeric	cancellation_rate numeric
1	SpiceRoute Mumbai 2	450	196557.00	44.26	3.75	7.34
2	SpiceRoute Delhi 1	467	199791.00	43.30	3.72	7.17
3	SpiceRoute Mumbai 1	432	191911.00	44.41	3.62	6.12
4	SpiceRoute Pune 2	438	200719.00	44.57	3.72	6.03
5	SpiceRoute Jaipur 1	464	210277.00	44.86	3.60	5.99
6	SpiceRoute Bengaluru 1	453	214089.00	43.42	3.69	5.87
7	SpiceRoute Pune 1	477	218320.00	43.90	3.81	5.78
8	SpiceRoute Bengaluru 2	457	210384.00	43.23	3.72	5.72
9	SpiceRoute Delhi 2	461	205496.00	44.69	3.71	5.18
10	SpiceRoute Jaipur 2	449	202519.00	43.83	3.74	5.18

KEY OBSERVATION

- Sales Performance: SpiceRoute Pune 1 generated the highest gross item sales.
- Operational Issues: SpiceRoute Mumbai 2 showed slower delivery and a higher cancellation rate.
- Customer Satisfaction: SpiceRoute Jaipur 1 received lower average ratings and may need service improvements.

BUSINESS QUESTION

Build a single report by outlet showing delivered orders, gross item sales, average delivery time, average review rating, and cancellation rate.

ANALYSIS OBJECTIVE

To compare outlet performance across sales, delivery efficiency, customer ratings, and cancellation rates.

SQL FILE

postgreSQL-15